BLUESTOCK MUTUAL FUND ANALYTICS

Capstone Project Report

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Project Title: Bluestock Mutual Fund Analytics

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Certificate

I hereby declare that the project titled **"Bluestock Mutual Fund Analytics"** is my original work carried out as part of my academic requirements. The work presented in this report has not been submitted previously for any other academic qualification.

Acknowledgement

I sincerely express my gratitude to Bluestock Fintech for providing me with the opportunity to work on this capstone project. Their guidance, support, and valuable feedback helped me successfully complete the project.

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1. Introduction

Mutual funds are one of the most preferred investment options for retail investors due to professional management and portfolio diversification. With thousands of schemes available in the market, analysing fund performance manually becomes difficult. This project aims to build a comprehensive Mutual Fund Analytics system that collects, cleans, analyses, and visualises mutual fund data. Using Python, SQL, and Power BI, the project provides meaningful insights into fund performance, investor behaviour, and market trends to support better investment decisions.

2. Objectives

- Collect and process mutual fund datasets.
- Clean and prepare the data for analysis.
- Store the processed data in an SQLite database.
- Perform exploratory data analysis (EDA).
- Calculate important performance metrics such as CAGR, Sharpe Ratio, Alpha, Beta, and Maximum Drawdown.
- Develop an interactive Power BI dashboard.
- Perform advanced analytics including Value at Risk (VaR), Rolling Volatility, Cohort Analysis, and Recommendation Logic.

3. Problem Statement

Investors often struggle to compare mutual funds because performance data is scattered across multiple sources and involves complex financial metrics. This project aims to centralise the data, automate performance calculations, and present meaningful insights through an interactive dashboard and analytical reports.

4. Dataset Description

The project uses multiple datasets related to the Indian mutual fund industry.

The datasets include:

- Fund Master
- NAV History
- AUM by Fund House
- Monthly SIP Inflows
- Category Inflows
- Industry Folio Count
- Scheme Performance
- Investor Transactions
- Portfolio Holdings
- Benchmark Indices

➢ These datasets were stored inside the **data/raw** directory and cleaned versions were saved inside **data/processed**.

5. Methodology

The project followed the following workflow:

1. Data Collection
2. Data Cleaning
3. Database Creation
4. Exploratory Data Analysis
5. Performance Analytics

6. Advanced Analytics
7. Dashboard Development
8. Documentation

The tools used include:

- Python
- Pandas
- NumPy
- Matplotlib
- SQLite
- SQL
- Power BI
- Git & GitHub

6. Data Cleaning

The data cleaning process involved:

- Removing duplicate records.
- Handling missing values.
- Standardising date formats.
- Correcting inconsistent values.
- Converting data types.
- Validating transaction records.
- Preparing cleaned datasets for analysis.
- ➢ The cleaned datasets were stored in the **data/processed** folder.

7. Exploratory Data Analysis

EDA was performed to understand patterns and trends in the data.

The analysis included:

- NAV Trends
- AUM Growth
- SIP Inflow Analysis
- Category Inflows
- Folio Growth
- Investor Demographics
- Geographic Distribution
- Correlation Analysis

➢ Visualisations such as line charts, bar charts, heatmaps, and pie charts were used to identify key insights.

8. Performance Analytics

The project calculated several financial performance metrics, including:

- Daily Returns
- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Tracking Error
- Maximum Drawdown

➢ These metrics were generated using Python scripts and saved as CSV files for further analysis.

9. Advanced Analytics

Advanced analytics extended the project by analysing investment risk and investor behaviour.

The following analyses were performed:

- Value at Risk (VaR)
- Rolling Volatility
- Rolling Sharpe Ratio
- Cohort Analysis
- Investor Segmentation
- Fund Recommendation System

➢ These analyses provide deeper insights into mutual fund performance and investment patterns.

10. Dashboard Overview

An interactive Power BI dashboard was developed consisting of five pages:

- Industry Overview
- Fund Performance
- Investor Analytics
- SIP & Market Trends
- Fund NAV Details

➢ The dashboard includes KPI cards, slicers, trend analysis, risk metrics, and interactive visualisations for better decision-making.

11. Key Findings

- Mutual fund AUM showed consistent growth during the analysis period.
- SIP inflows increased steadily, reflecting growing investor confidence.
- Large-cap funds generally exhibited more stable performance than higher-risk categories.

- Risk-adjusted metrics such as the Sharpe Ratio helped identify better-performing funds.
- Cohort analysis highlighted investor retention trends.
- The recommendation model classified funds based on performance and risk metrics.

12. Conclusion

The Bluestock Mutual Fund Analytics project successfully demonstrated the end-to-end process of analysing mutual fund data using modern data analytics tools. The project integrated data collection, cleaning, database management, financial metric calculations, advanced analytics, and interactive visualisation into a single workflow. The resulting dashboard and analytical reports provide valuable insights that can support investors in making more informed investment decisions.

13. Future Scope

Future improvements may include:

- Real-time NAV updates.
- Streamlit web application deployment.
- Portfolio optimisation using Markowitz Efficient Frontier.
- Monte Carlo simulation for future NAV forecasting.
- Automated email reporting.
- Machine learning-based recommendation systems.

14. References

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5. Pandas Documentation

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7. SQLite Documentation
8. Matplotlib Documentation
9. Bluestock Learning Resources